Artificial Intelligence Assignment 3: Clustering

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September 24, 2025

Contents

1 Clustering Problems

1.1 K-Means Clustering

1.1.1 Conceptual Overview

K-Means is an iterative algorithm that partitions a dataset into a pre-defined number (k) of distinct clusters. It works in two main steps:

1. **Assignment Step:** Each data point is assigned to the cluster with the nearest center (centroid).

2. **Update Step:** The centroid of each cluster is recalculated as the mean of all data points assigned to it.

These steps repeat until cluster assignments no longer change (convergence).

1.1.2 Question 1

Using the k-means algorithm, place the following points into three clusters. Assume the initial cluster centers are the points A1, A4, and A7, respectively. Run the algorithm for only one epoch.

Points: $A1 = (2, 10)$, $A2 = (2, 5)$, $A3 = (8, 4)$, $A4 = (5, 8)$, $A5 = (7, 5)$, $A6 = (6, 4)$, $A7 = (1, 2)$, $A8 = (4, 9)$.

(a) After one epoch, specify the cluster for each point. We calculate the Euclidean distance of each point to the three initial centroids: $C1 = (2, 10)$, $C2 = (5, 8)$, $C3 = (1, 2)$.

- **A1(2,10)**: $d(A1, C1) = 0$. Belongs to **Cluster 1**.

- **A2(2,5)**: $d(A2, C1) = 5, d(A2, C2) = \sqrt{18} \approx 4.24, d(A2, C3) = \sqrt{10} \approx 3.16$. Belongs to **Cluster 3**.

- **A3(8,4)**: $d(A3, C1) = \sqrt{72} \approx 8.48, d(A3, C2) = 5, d(A3, C3) = \sqrt{53} \approx 7.28$. Belongs to **Cluster 2**.

- **A4(5,8)**: $d(A4, C2) = 0$. Belongs to **Cluster 2**.

- **A5(7,5)**: $d(A5, C1) = \sqrt{50} \approx 7.07, d(A5, C2) = \sqrt{13} \approx 3.61, d(A5, C3) = \sqrt{45} \approx 6.71$. Belongs to **Cluster 2**.

- **A6(6,4)**: $d(A6, C1) = \sqrt{52} \approx 7.21, d(A6, C2) = \sqrt{17} \approx 4.12, d(A6, C3) = \sqrt{29} \approx 5.38$. Belongs to **Cluster 2**.

- **A7(1,2)**: $d(A7, C3) = 0$. Belongs to **Cluster 3**.

- **A8(4,9)**: $d(A8, C1) = \sqrt{5} \approx 2.24, d(A8, C2) = \sqrt{2} \approx 1.41, d(A8, C3) = \sqrt{58} \approx 7.61$. Belongs to **Cluster 2**.

(b) Calculate the new centers.

- **Cluster 1:** $\{A1\}$. New $C1 = (\mathbf{2}, \mathbf{10})$.
- **Cluster 2:** $\{A3, A4, A5, A6, A8\}$. New $C2 = (\frac{8+5+7+6+4}{5}, \frac{4+8+5+4+9}{5}) = (\mathbf{6}, \mathbf{6})$.
- **Cluster 3:** $\{A2, A7\}$. New $C3 = (\frac{2+1}{2}, \frac{5+2}{2}) = (\mathbf{1.5}, \mathbf{3.5})$.

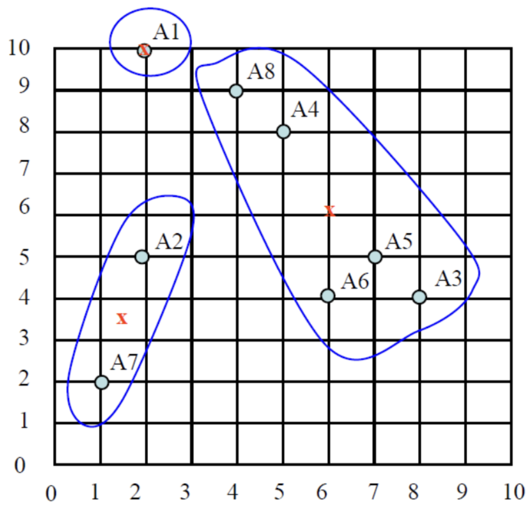

Figure 1: Clusters after Epoch 1. Red 'x' marks the new centroid.

(c) Plot the points and their clusters.

(d) How many more iterations are needed for the algorithm to converge? For each iteration, simply draw the points and their clusters.

Answer: Two more iterations are needed. The algorithm converges after the 3rd epoch in total.

- **After Epoch 2:** Clusters are $\{A1, A8\}$, $\{A3, A4, A5, A6\}$, $\{A2, A7\}$. New centroids are $C1 = (3, 9.5)$, $C2 = (6.5, 5.25)$, $C3 = (1.5, 3.5)$.

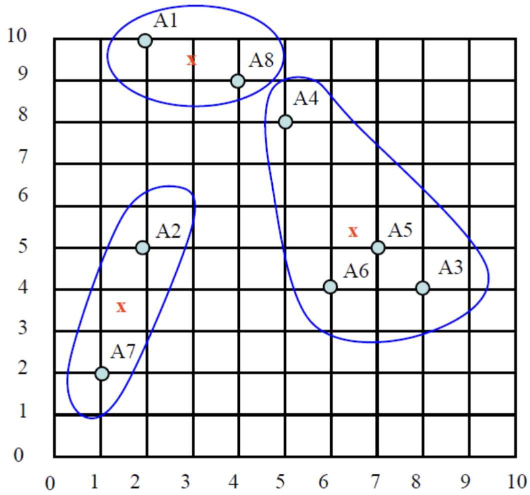

Figure 2: Clusters after Epoch 2.

- **After Epoch 3 (Converged):** Clusters are $\{A1, A4, A8\}$, $\{A3, A5, A6\}$, $\{A2, A7\}$. New centroids are $C1 = (3.66, 9)$, $C2 = (7, 4.33)$, $C3 = (1.5, 3.5)$. The assignments do not change in the next iteration.

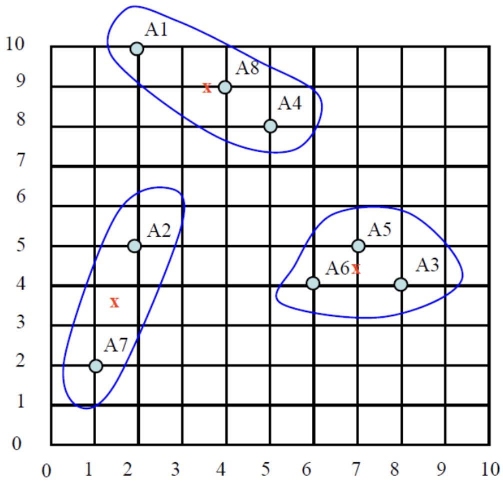

Figure 3: Clusters after Epoch 3 (Convergence).

1.2 DBSCAN Clustering

1.2.1 Conceptual Overview

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) groups together points that are closely packed, marking as outliers (noise) points that lie alone in low-density regions. It uses two parameters:

- **Epsilon (ϵ):** The radius of the neighborhood around a point.

- **minPoints:** The minimum number of points required within the ϵ-neighborhood to form a dense region.

Points are classified as **core**, **border**, or **noise**.

1.2.2 Question 2

Cluster the points from the previous question with the DBSCAN algorithm. First consider $\epsilon = 2$ and then $\epsilon = \sqrt{10}$. In both cases, consider $minPoints = 2$. Draw the shape of the points and clusters. Also, specify the neighboring points of each point and the noise points.

Case 1: $\epsilon = 2$, $minPoints = 2$

- **Neighbors:** A point is a neighbor if its distance is ≤ 2. The neighbors for each point are:
  - A1: -
  - A2: -
  - A3: {A5, A6}
  - A4: {A8}
  - A5: {A3, A6}
  - A6: {A3, A5}
  - A7: -
  - A8: {A4}
- **Noise Points:** A1, A2, and A7 have fewer than $minPoints - 1 = 1$ neighbor, so they are noise.
- **Clusters:**
  - Cluster 1: {A3, A5, A6} (density-connected).
  - Cluster 2: {A4, A8} (density-connected).

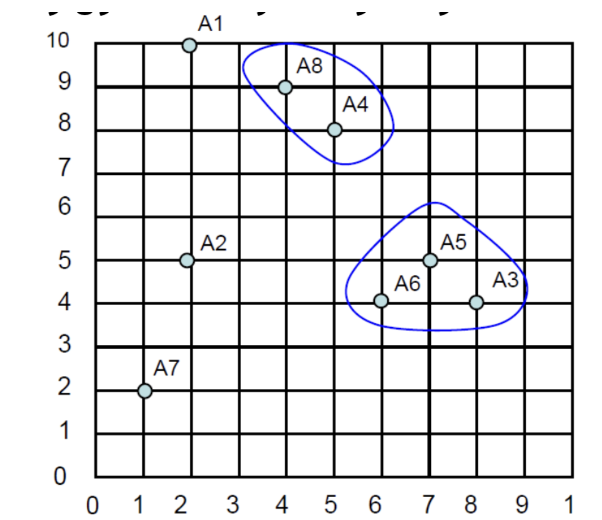

Figure 4: DBSCAN clusters for $\epsilon = 2$. A1, A2, A7 are noise.

Case 2: $\epsilon = \sqrt{10} \approx 3.16$, $minPoints = 2$

- **Neighbors:**
  - A1: {A8}
  - A2: {A7}
  - A3: {A5, A6}
  - A4: {A8}
  - A5: {A3, A6}
  - A6: {A3, A5}
  - A7: {A2}
  - A8: {A1, A4}
- **Noise Points:** None. Every point has at least one neighbor, so all points are core points.
- **Clusters:**
  - Cluster 1: {A1, A4, A8}.
  - Cluster 2: {A2, A7}.
  - Cluster 3: {A3, A5, A6}.

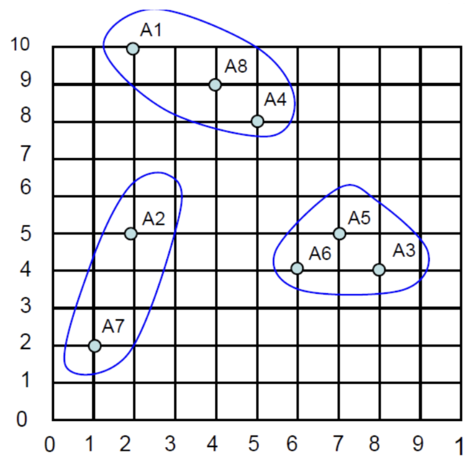

Figure 5: DBSCAN clusters for $\epsilon = \sqrt{10}$. No noise points.

1.3 Agglomerative Hierarchical Clustering

1.3.1 Conceptual Overview

Agglomerative Clustering is a "bottom-up" hierarchical method. It starts with each point as its own cluster and progressively merges the closest pair of clusters. The distance between clusters is defined by a **linkage criterion**. Here, we use **single linkage**, where the distance between two clusters is the minimum distance between any two points in the respective clusters. The process is visualized with a **dendrogram**.

1.3.2 Question 3

Using the agglomerative clustering algorithm and the distance matrix below, cluster points A to D step-by-step and draw their hierarchical diagram. Consider the distance between two clusters to be the distance of their closest points.

	A	**B**	**C**	**D**
A	0	1	4	5
B	1	0	3	6
C	4	3	0	2
D	5	6	2	0

Step-by-Step Clustering

Step	Distance	Clusters	Explanation
0	0	{A}, {B}, {C}, {D}	Initially, each point is its own cluster.
1	1	{A, B}, {C}, {D}	Since the two clusters A and B are the closest and their distance is one, they are combined.
2	2	{A, B}, {C, D}	Since the two clusters C and D are the closest and their distance is two, they are combined.
3	3	{A, B, C, D}	Finally, the remaining two clusters, whose distance is three, are combined, and one cluster remains.

Dendrogram The dendrogram shows the merges at heights corresponding to the merge distances.

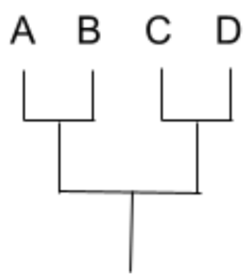

Figure 6: Dendrogram showing the hierarchical clustering.